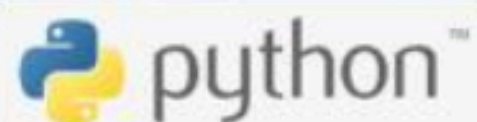


ONLINE HACKATHON: RETAIL **CRISIS** & **RECOVERY** Visualization Challenge



CRISIS START

RECOVERY

RECOVERY

LOWEST
POINT

#HACK-2026-PYTHON-01

By : Nabila Hulwana Z

Problem Background

```
graph TD; A[Problem Background] -.-> B((01)); A -.-> C((02)); A -.-> D((03)); A -.-> E((04)); A -.-> F((05)); B --- G[DQFresh Mart: a retail store with one branch]; C --- H[Last 6 months: total sales continuously declining]; D --- I[Last 6 months: total sales continuously declining]; E --- J[Problem: some small SKUs are growing consistently but not detected by the system]; F --- K[Frequently out of stock -> went unnoticed by cashiers];
```

01

**DQFresh Mart: a
retail store with
one branch**

02

**Last 6 months:
total sales
continuously
declining**

03

**Last 6 months:
total sales
continuously
declining**

04

**Problem: some
small SKUs are
growing
consistently but
not detected by
the system**

05

**Frequently out
of stock → went
unnoticed by
cashiers**

Analysis Objective

01

Identify Rising Star, small products that are growing consistently

02

Find co-purchase patterns for bundling strategy

03

Generate relative growth vs actual visualizations

Dataset

Column - Description

- 1.nomor_struk — Transaction invoice ID
- 2.tgl_transaksi — Transaction date
- 3.kode_produk — SKU code
- 4.nama_produk — Product name
- 5.jumlah_terjual — Quantity sold
- 6.harga — Unit price
- 7.total_nilai — price × quantity

	A	B	C	D	E	F	G	H
1	nomor_struk	tgl_transaksi	kode_produk	nama_produk	jumlah_terjual	harga	total_nilai	
2	INV-00000001	01-02-2025	MARGRN	Margarin Serbaguna	4	18000	72000	
3	INV-00000002	01-02-2025	DISFIL	Cairan Disinfektan 1L	5	60500	302500	
4	INV-00000002	01-02-2025	AMGALN	Air Mineral Galon	1	15500	15500	
5	INV-00000002	01-02-2025	SNKKPT	Snack Keripik Kentang	1	20500	20500	
6	INV-00000002	01-02-2025	BANTAL	Bantal Tidur Silikon	2	66500	133000	
7	INV-00000002	01-02-2025	KSKK3P	Kaos Kaki (3 Pasang)	2	25000	50000	
8	INV-00000002	01-02-2025	TEKOLT	Teko Listrik	2	89000	178000	
9	INV-00000002	01-02-2025	KBLCHG	Kabel Data Fast Charge	1	59000	59000	
10	INV-00000002	01-02-2025	SACTAN	Sabun Cuci Tangan Refill	1	15000	15000	
11	INV-00000003	01-02-2025	SMBAT3	Sabun Mandi Batang (Isi 3)	1	11000	11000	
12	INV-00000003	01-02-2025	KSKK3P	Kaos Kaki (3 Pasang)	1	25000	25000	
13	INV-00000004	01-02-2025	TRMS50	Termos Air Panas 500ml	3	81000	243000	
14	INV-00000004	01-02-2025	KBLCHG	Kabel Data Fast Charge	8	59000	472000	
15	INV-00000004	01-02-2025	SHP5AC	Shampo Sachet	1	17500	17500	
16	INV-00000005	01-02-2025	KSKK3P	Kaos Kaki (3 Pasang)	2	25000	50000	
17	INV-00000006	01-02-2025	TEKOLT	Teko Listrik	2	89000	178000	
18	INV-00000006	01-02-2025	KSKK3P	Kaos Kaki (3 Pasang)	3	25000	75000	
19	INV-00000006	01-02-2025	GRMDAP	Garam Dapur (3 Pack)	3	10500	31500	
20	INV-00000006	01-02-2025	MSKMED	Masker Medis (1 Box)	1	52500	52500	
21	INV-00000006	01-02-2025	TLR05K	Telur Ayam (Serangap Ka)	1	16500	16500	

Metodology : Rising Star

How to Find Rising Star Products

01

Smoothing:

3-day Moving Average per product. Data is aggregated by day per product first before MA is calculated.

$MA(t) = \text{rata-rata total_nilai selama 3 hari terakhir}$

02

Trend Detection:

A product is in a rising trend if $MA(\text{today}) > MA(\text{yesterday})$

03

Filter:

Only products with ≥ 12 consecutive days in a rising trend

04

Growth %:

average of total_nilai over the last 3 days

$\text{Growth \%} = ((MA_{\text{end}} - MA_{\text{start}}) / MA_{\text{start}}) \times 100$

Metodology : Potential Packaging

Association Rules with the Apriori Algorithm

01

Basket matrix:

pivot of transactions per
receipt × product (binary 0/1)

02

min_support:

0.01 (1% from total transaction)

03

metric: lift, min_threshold: 1

04

Final Filter :

- at least one item in the rule must be a Rising Star
- Lift ≥ 2

05

Sort Order :

Lift → Support → Confidence
(descending)

Visualization: Rising Star Index

Relative Growth_(Base 100).

01

All products normalized starting from 100

02

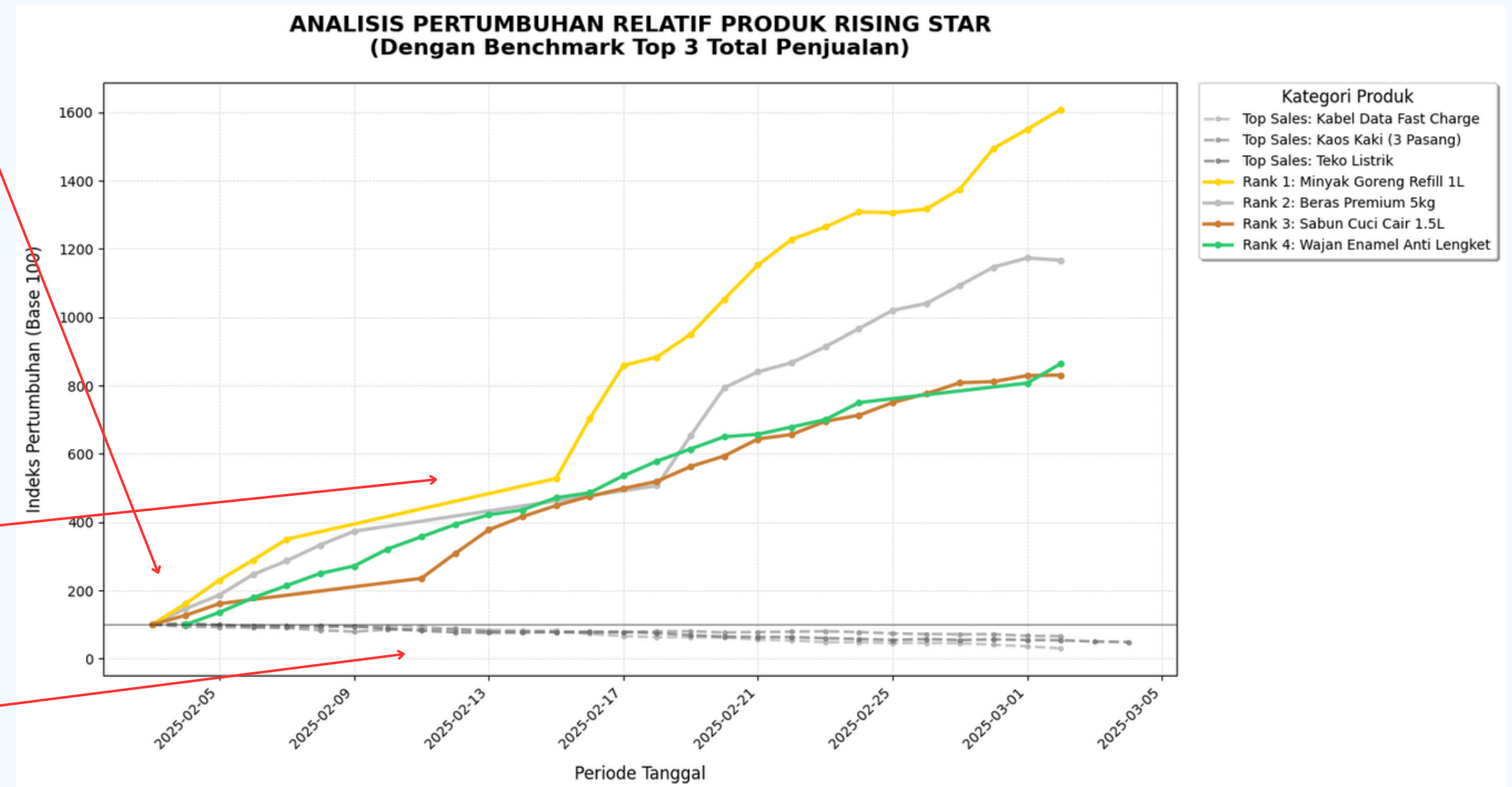
Compares growth speed of Rising Star vs Top 3 Sales

03

Rising Star: colored lines (gold, silver, bronze, etc.)

04

Top 3 Sales: gray dashed lines



Visualization: Actual Value

Original Sales Value

01

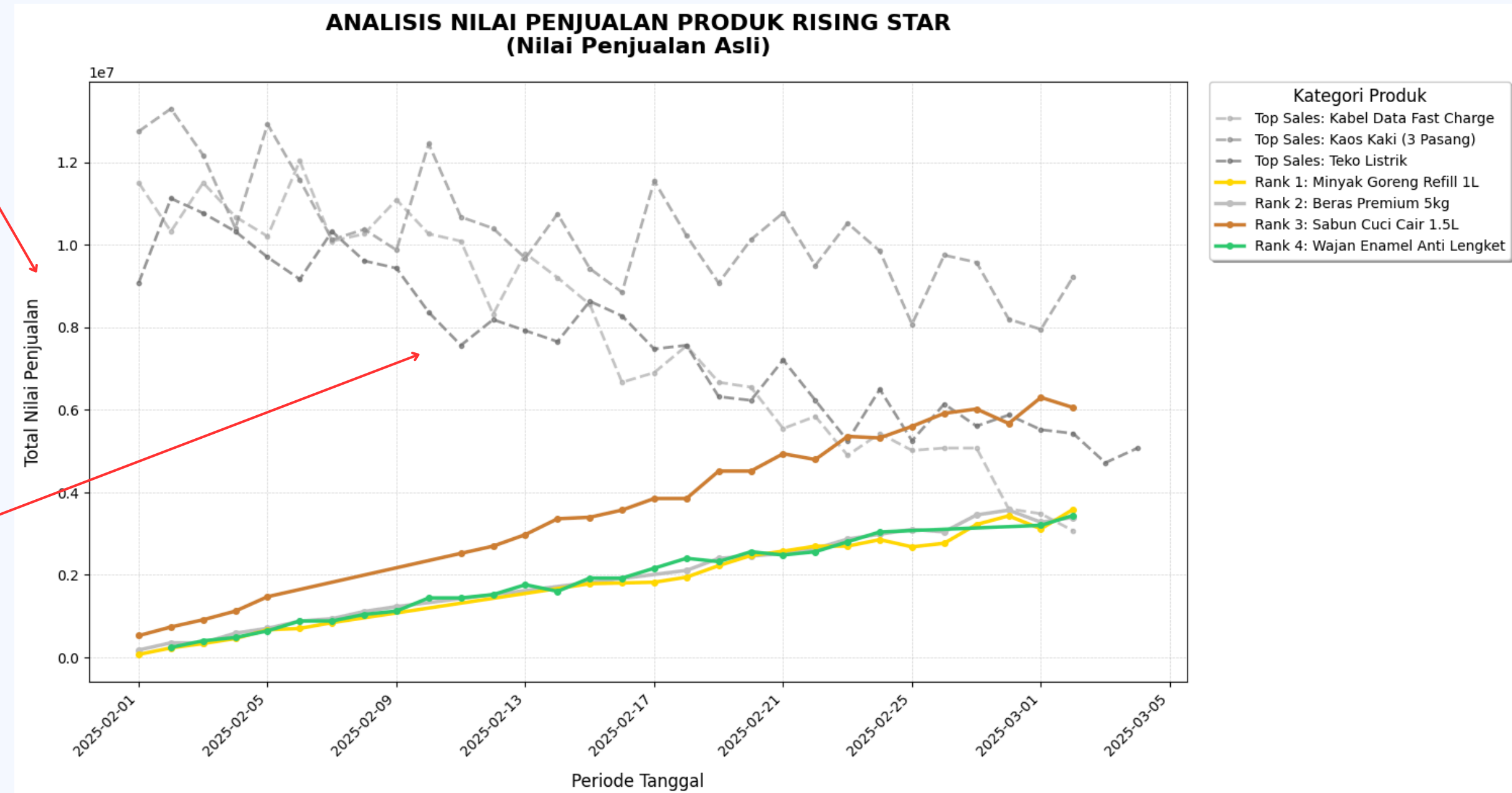
Displays daily total_nilai per product (without normalization)

02

Demonstrates why these products aren't detected traditionally (Top 3 Sales have higher values than Rising Star)

03

Context: Rising Star appears "small" compared to Top 3 Sales



Findings & Follow-up: retail_insight.xlsx

2 sheet: Rising Star + Potential Packaging

01

Rising Star Products → restock immediately

02

Potential Packaging → create bundling packages based on association rules

03

High lift = strength of co-purchase association

04

Strategy: cross-selling, package promos, placing products close together on display

	A	B	C	D
1	Kode Produk	Nama Produk	Growth %	Total Penjualan
2	MGR1L	Minyak Goreng Refill 1L	712.07	44,940,000
3	BRS5KG	Beras Premium 5kg	700.00	44,109,000
4	SCC15L	Sabun Cuci Cair 1.5L	551.90	95,970,000
5	WJANEM	Wajan Enamel Anti Lengket	536.84	44,240,000
6				
7				

	A	B	C	D	E	F
1	Jika Membeli	Maka Membeli	Jumlah Invoice	Support	Confidence	Lift
2	Beras Premium 5kg	Minyak Goreng Refill 1L	100	0.01	0.26	2.3
3	Minyak Goreng Refill 1L	Beras Premium 5kg	100	0.01	0.1	2.3
4	Sabun Cuci Cair 1.5L, Kaos Kaki	Minyak Goreng Refill 1L	185	0.02	0.25	2.26
5	Minyak Goreng Refill 1L	Sabun Cuci Cair 1.5L, Kaos Kaki	185	0.02	0.18	2.26
6	Minyak Goreng Refill 1L, Kaos Kaki	Sabun Cuci Cair 1.5L	185	0.02	0.21	2.23
7	Sabun Cuci Cair 1.5L	Minyak Goreng Refill 1L, Kaos Kaki	185	0.02	0.2	2.23
8	Kaos Kaki (3 Pasang), Cairan Pembersih	Sabun Cuci Cair 1.5L	103	0.01	0.2	2.11
9	Sabun Cuci Cair 1.5L	Kaos Kaki (3 Pasang), Cairan Pembersih	103	0.01	0.11	2.11
10	Sabun Cuci Tangan Reffil, Minyak Goreng	Mouse Wireless, Kaos Kaki	98	0.01	0.22	2.1

Tech Stack

Tools & Library used



01

Python
3.10 - 3.14

for Runtime



02

pandas
2.3.1

Data
Processing,
MA,
Filtering



03

matplotlib
3.10.7

Chart
visualization



04

mlxtend
0.23.4

Algoritma
Apriori &
Association
Rules



05

openpyxl
3.1.5

Excel export &
Formatting

Strategic Recommendation

01

"Star-Package" Bundling

Create product packages combining Rising Star products with their paired products based on Apriori results.

Goal: increase average transaction value

02

Momentum-Based Stock Prioritization

Add safety stock for Rising Star products. A long streak = real and sustained demand.

Goal: avoid stockouts during peak momentum

03

Campaign Focused on Proven Products

Focus promotional efforts and marketing budget on the 18 Rising Star products, not on declining products.

Goal: strengthen what's already working, don't waste budget trying to revive products that are already dying



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training

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.com

CERTIFICATE of Achievement

Is presented to:

Nabila Hulwana

for achieving a position in the Top 100 Finalist

Rank #88 out of 464 Participants

In the **Hackathon: Retail Crisis & Recovery Visualization Challenge using Python**

9 May 2026

Selected among top-performing participants for demonstrating exceptional mastery in building end to end data pipeline, data analysis, dashboard development, and strategic insights using Python.

#DQLABHCK100V2UKVSEQ

Yovita Surianto
Manager DQLab

Feris Thia
Founder Xeratic



Thank You!

**DQLab × UjiKompetensi
Hackathon 2026**

Lets Connect :

Linkedin : [nabila hulwana](#)

github : [dqlab python hackathon](#)